

for different customers. The exchange has its own private keys to the cryptoassets it is responsible for on the respective blockchain and then has internal books that record the customer balances. Depending on the exchange, there are varying levels of security hygiene and different proportions of the exchange's assets that are kept in hot or cold storage. Over time, these security distinctions have proven critical. For a clearer understanding, we'll cover a few big hacks that occurred on bitcoin exchanges that stored 100 percent of their bitcoin in hot wallets.

Let's begin with the infamous Mt. Gox. While this exchange did much to expand the usage and recognition of bitcoin throughout the world, it met its end in early 2014²⁵ when the company declared bankruptcy after over \$450 million²⁶ of client bitcoin holdings went missing. Although the company was a pioneer in providing investors and enthusiasts the opportunity to gain access more easily to bitcoin, Mt. Gox also had weak management involved in an asset class that was still in its infancy—never a good combination.

Jed McCaleb was the original owner of Mt. Gox. Early on he learned that matching buyers and sellers of bitcoin was more than he bargained for when wires for tens of thousands of dollars started to pour in. McCaleb sold the site, and its growing activity, to Mark Karpeles, who was known in chat rooms as MagicalTux and enjoyed posting kitten videos online. To his credit, Karpeles rewrote the site to address the increased interest and activity, and he survived through the early days when other bitcoin exchanges quickly folded.²⁷